Biology by the Numbers Spring 2026: Lecture 1

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Abstract. Notes for teaching Biology By The Numbers (BTN) in Spring 2026 at Rockefeller University.

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1 Lecture 1 – setting the tone, syllabus review

Intros around the room. Name, why you took the class, what you're excited about scientifically.

2 Why are we here

Some suggestions: Understanding the living world, learning to use numbers thoughtfully. Numbers are not decorations. Rather, an ecosystem of relationships that you need to develop the practice of using and mastering.

You are invited to join me in "street fighting" calculations in biology

First lecture covers:

1. Methodology – active learning
2. Some rules of thumb
3. Divide and conquer approach
4. Simple math, big effect
5. All the math you need is from middle school

You might feel some knee jerk antagonism to some of the things we do, e.g. $5 \approx 10$ and 'a few hundred'. Try to bear with me and I hope you will se

Outline for Board

1. Course in 2026
2. Why are we here?
3. Concrete learning
4. Sixth sense
5. Fermi
6. Olympics
7. Divide and conquer
8. Few to many
9. New questions
10. Number proteins
11. Syndrome
12. Math
13. Coli
14. Units
15. 1, few, 10
16. Answer
17. Generalizations
18. Expectations
19. Attendance
20. Grading
21. Teaser
22. My notes will be available

3 What are we hoping to get out of this?

Suggestions from the room?

Calling BS is very important in 2026. We are bombarded with information / ideas, and many of them are very low quality. Social media and AI magnify the problem.

Some ideas:

1. A way of thinking (*could get from youtube / MOOC too*)
2. Set aside time to learn
3. Interactive: answer questions, ask questions, think on your feet
4. Know your peers, learn from them,
5. Learning together is different — presentation skills, pairs, reverse classroom
6. Upgrade your research — gain skills relevant to research, engage peers
7. Enjoy

The course will try to take you from looking at numbers as "decorations" that may support the qualitative deductions to things that can have value and meaning by themselves.

the main thing it gave me is a motivation to actually LOOK at numbers presented in papers. Before, I used to sort of ignore the actual number [and just look at the effect,] but now I actually think about the number itself and its meaning relative to other things.

Mor Kenigsbuch, WIS BTN 2017-8

3.1 Our approach

Our main method is case studies — worked example calculations. I will do these on the board and you will do them in your assignments and projects. Will share examples from real world. If you were not active (in the class, projects, assignments) something went wrong.

Our goals are to

1. Gain facility with the order-of-magnitude (OOM) approach
2. Learn useful biological numbers applicable to biological research
3. Become quick with calculations
4. See connections between scales and fields
5. Gain an "additional sense" –

We want to gain a feeling for the numbers in biology. Often we talk about a feeling for the organism (McClintock book) but we can only have a feel for a few aspects of a few organisms. Numeracy is a complementary perspective that we hope to give you here.

4 MOOCs, numbers as decorations

An email from a MOOC provider to one of the book authors, Ron Milo.

> Dear Ron Milo,
> As we near the end of 2016 and look to the New Year, I want to say a heartfelt thank you for being a part of the global edX learner community. Together, we have accomplished so much – 33,601,092 course enrollments, 117,969,427 problem sets attempted, 229,405,155 minutes of video watched in more than 1,300 courses from the world's best universities and institutions.

What should we make of these numbers? Suggestions?

Minutes per enrollment: 230 M / 34 M ≈ 10 minutes per course.
Talk about 229/33.6 = 6.82 vs 230 / 34 ≈ 10.

Enrollements per course: 34 M enrollments / 1300 courses $\approx 3 \times 10^3$ enrollments/course. *A regular course might have 10-100 students*

Minutes per course: 230 M minutes / 1300 courses $\approx 2 \times 10^5$ minutes / course or ≈ 3000 hrs / course over all the students
Do with powers of 10, emphasize OOM approach.

How many students is this? We could compare to our course.
≈ 10 lectures x 100 min / lecture $\approx 10^3$ min / student
So the online courses have ≈ 200 "real students" students effectively. *could include the homework effort and get a larger number*

Mention 6th sense here...

See handout with quotes at end...

5 Fermi problems – inference with no data

How many piano tuners in Chicago?

Break it down into steps. How many people in Chicago? What fraction have a

piano? How often do they have to be tuned? Can work your way to an estimate without data.

Suggestions for OOM questions you've heard of.

Mention these kinds of brain teasers used in job interviews. Very common in engineering and physics – called "back of the envelope" calculations.

Some examples:

1. how many windows in NYC?
2. how many buckets of paint does NYC use to paint roads each year?
3. estimate the number of iphone apps downloaded each day
4. how many electric cars are sold each year?
5. how many golf balls fit in a schoolbus?

For these we have some intuition from our lives

In biology we often lack the direct experience (the kind used to do the piano tuner calc). We are left with logic and a few "rules of thumb." You'll see we can get far with just a few numbers.

Back to Chicago:
Outline: (# pianos) x (tunings / piano / yr) / (# tunings / tuner / yr)
Pianos: $f \times 10^6$ people $\rightarrow$ 1:10 to 1:100 play the piano $\rightarrow$ $f \times 10^5$ pianos.
Tuners: tune f/day, 200 days/year, ≈1000 pianos/year.
Say a piano has to be tuned every f years.
So every tuner is in charge of $f \times 10^3$ pianos.
$f \times 10^5$ pianos / ($f \times 10^3$ pianos/tuner) $\approx$ **100 tuners.**

6 Olympics – 100 m dash

How long does it take to run the 100 m? World record by Usain Bolt (2009) is 9.58 s ≈ 10 s.

The olympic judges want to know: how fast could a sprinter possibly respond to the starting gun. *At what response time should we disqualify an experienced sprinter?*
a. <1ms b. <10ms c. <100ms d. <1000ms.

Take 5 minutes to think of a strategy or even answer in pairs. Try without internet.

Strategy – "divide and conquer": Sketch the approach with a drawing of a sprinter A (shot to ear); B (ear and brain); C(brain to leg); (D; neuromuscular order to actual movement))

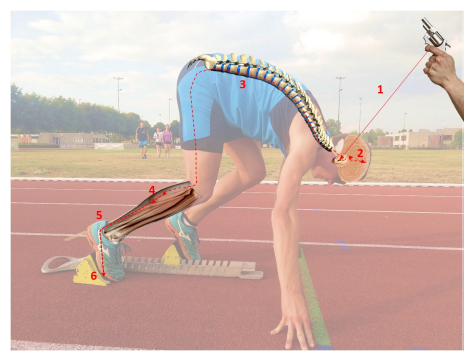

Figure 1: A sprinter at the starting block. See a fuller calculation from Bas Van Hooren.

What is relevant? Speed of sound 300 m/s. Calculate, $\approx f$ m/300m/s $\approx f \times 10^{-3}$ ms mention – in the Olympics they moved to electronic loudspeakers in starter blocks.

Action potential. Need 1-2 m of conduction. Speed 10-100 m/s.
≈ 2 m/$(10 - 100$ m / s) $\approx 20 - 200$ ms

We see the power of rules of thumb from this. Don't need to know all the details of neurons for all calculations. A few numbers give us the ability to calculate many things.

Olympic regulations: anything below 100 ms is disqualified (based on pressure gauges).

New meaning to responding in split second.

Any reactions here?

Some thoughts:

1. Perhaps the muscles processes are slow?
2. Maybe you can train for this.

Turns out that folks who trained can do it in ≈ 100 ms.

This discussion brings new questions: 24 frames per second is the standard frame rate for TV and movies. Just enough to see 100 ms reactions. High speed cameras can be much faster (10^3 fps or faster).

Once can think about the response time of muscle, or the movement of myosin motors on actin.

One can also think about the time time for a neurotransmitter to traverse the synaptic junction. The junction has width of 10-100 nm [BNID 112017]. Say it's 100 nm and D=1000 um2/s. If we treat this as 1D diffusion (we'll touch on this later), then

$$t_{diff} = \sqrt{\frac{r^2}{2D}} = \sqrt{\frac{10^{-2}}{2 \times 10^3}}$$
$$= \sqrt{1/2 \times 10^5} \approx \frac{1}{5 \times 10^2} = 2 \times 10^{-3} \text{ s}$$

or 2 milliseconds.

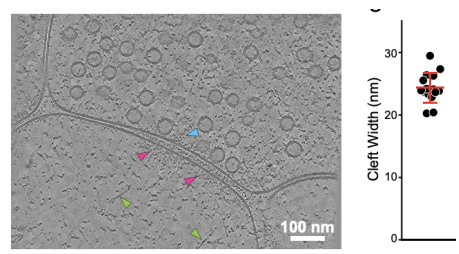

Figure 2: Size of mouse synapses, Held et al. PNAS 2024.

from Wikipedia on False start

An analysis of start times by sprinters at the 2008 Beijing Olympics demonstrated that male and female sprinters can achieve reaction times of 109 and 121 ms in one out of 1,000 starts.

from Wikipedia on 100 m dash

At high level meets, the time between the gun and first kick against the starting block is measured electronically, via sensors built in the gun and the blocks. A reaction time less than 0.1 s is considered a false start. The 0.1-second interval accounts for the sum of the time it takes for the sound of the starter's pistol to reach the runners' ears, and the time it takes to react to it. For many years a sprinter was disqualified if responsible for two false starts individually. However, this rule allowed some major races to be restarted so many times that the sprinters started to lose focus. The next iteration of the rule, introduced in February 2003, meant that one false start was allowed among the field, but anyone responsible for a subsequent false start was disqualified. This rule led to some sprinters deliberately false-starting to gain a psychological advantage: an individual with a slower reaction time might false-start, forcing the faster starters to wait and be sure of hearing the gun for the subsequent start, thereby losing some of their advantage. To avoid such abuse and to improve spectator enjoyment, the IAAF implemented a further change in the 2010 season – a false starting athlete now receives immediate disqualification.[4] This proposal was met with objections when first raised in 2005, on the grounds that it would not leave any room for innocent mistakes. Justin Gatlin commented, "Just a flinch or a leg cramp could cost you a year's worth of work."[5] The rule had a dramatic impact at the 2011 world championships, when current world record holder Usain Bolt was disqualified.[6][7]

How many proteins are in a bacterial cell?

Possible answers:

1. $10^3 - 10^5$
2. $10^5 - 10^7$
3. $10^7 - 10^9$
4. $10^9 - 10^{11}$

We have no intuition about being a bacterium — and these are really big numbers, so we have no intuition about them either. Our approach is "divide and conquer", as we saw with the starting gun example.

Take 5 minutes to think of a strategy, or even an answer.

Mention impostor syndrome?

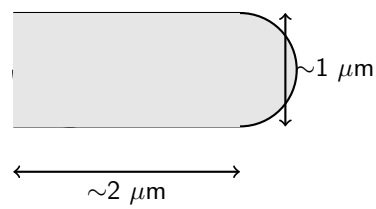

Figure 3: Geometry of *E. coli* cells made with the help of Claude.

Strategy: divide and conquer

Cell size $\rightarrow$ cell mass $\rightarrow$ mass of proteins in cell $\rightarrow$ number of proteins in cell

There are lots of ways to do this — there is value in seeing multiple approaches.

Cell size and mass

$$V_{\text{cell}} \approx (1\,\mu\text{m})^3 = \left(10^{-6}\,\text{m}\right)^3 = 10^{-18}\,\text{m}^3$$

$$M_{\text{cell}} = V_{\text{cell}} \times \rho = 10^{-18}\,\text{m}^3 \times 10^3\,\frac{\text{kg}}{\text{m}^3} = 10^{-15}\,\text{kg} = 10^{-12}\,\text{g}$$

(= 1pg = 1000 fg)

What is the mass of a protein? (Ask the class). If you are a biochemist, you know these numbers in units of kDa.

The Dalton and Avogadro's number

A **Dalton** (Da) is the mass of one hydrogen atom — our bridge between the macroscopic and microscopic worlds. Moving between these scales challenges our intuition; it helps to work with explicit numbers.

N_A is defined as the number of hydrogen atoms in 1 gram:

$$N_A \cdot \text{Da} = 1\,\text{g}, \qquad N_A = 6.02 \times 10^{23} \approx 10^{24}$$

How many Da in a cell?

$$10^{-12}\ \text{g/cell} \times \frac{10^{24}\ \text{Da}}{1\ \text{g}} = 10^{12}\text{Da/cell}$$

Sanity check of Da/g. Treat a hydrogen atom as a cube of side $1\,\text{Å}$, and use this to derive the number of Da per gram:

$$\frac{1\,\text{Da}}{\text{Å}^3} \times \left(\frac{10^8\,\text{Å}}{\text{cm}}\right)^3 = 10^{24}\,\frac{\text{Da}}{\text{cm}^3}$$

$$10^{24}\,\frac{\text{Da}}{\text{cm}^3} \times \frac{1\,\text{cm}^3}{\text{g}} = 10^{24}\,\frac{\text{Da}}{\text{g}} \approx N_A$$

Protein content of a cell

A bacterial cell is mostly water ($\sim 70\%$); roughly half the remaining dry mass is protein, giving $\sim 15\%$ protein by mass:

$$M_{p,\text{cell}} = M_{\text{cell}} \times 0.15 \ \sim \ 10^{11} \text{ Da of protein}$$

A typical protein has mass $m_p \sim 30$ kDa (from: ~ 300 amino acids $\times$ 100 Da/aa), so the number of proteins per cell is:

$$N_p = \frac{10^{11} \text{ Da}}{3 \times 10^4 \text{ Da/protein}} = 3 \times 10^6 \ \frac{\text{proteins}}{\text{bacterial cell}}$$

This is already quite useful! We learned about the number of proteins in E. coli, but it turns out that this volumetric number density is about right also for yeast and human cells.

Rule of Thumb: Protein molecules per μL

$\approx 2 - 4 \times 10^6$ proteins / μL [a]

[a] See Milo BioEssays 2013

This rule of thumb is potentially also useful for plant cells, but you would need to account for vacuole. We could make an estimate for plant cells, but it would force us to learn a bit about plant biology, e.g., volume fraction and contents of the vacuole, which organisms and cell types have one, etc.

Does this provoke any thoughts? Perhaps there is a physical limit here? Because of how we derived it, we see there would need to be (a) less water, (b) less other stuff (lipid, RNA, etc) or (c) smaller proteins for there to be more proteins / volume. Provokes the idea of physical limitations, but we'd need to do more work.

As a bonus, we can infer the number of mRNA molecules using a protein-to-mRNA ratio of $\sim 1000 : 1$, giving ~ 3000 mRNA per μL. This number is about right for E. coli, but doesn't work as well for yeast or human cells, which provokes us to think about differences in these organisms regulatory architectures.

7 Grading & Expectations

Class is pass / fail. Our expectations are that you will come to class, do the assignments and project genuinely and with effort. If you do all this you will pass. Simple as that.

You will work hard. We are going to work hard to help you gain the skills of quantitative thinking.

7.0.1 Attendance

Attendance at all lectures is **required**. If you cannot attend, send an email to the course address. You will need to read the lecture notes and write up the coolest thing you learned (or a cool thing we should have said).

Lectures from prior courses are on YouTube — better to watch after the coming to the in person lecture.

Rationale:

- We want to give you a useful skill. It comes from doing, not watching.
- Instead of a test with condensed preparation, we use continuous learning — this makes it deeper.

7.1 Assignments

There will be 4 assignments. The intention is that you do them on your own. If you fought with it and are stuck – go ahead and talk to your peers. Keep it in that spirit. Try not to use LLMs – this is not a good way to improve your capabilities.

Mention bonus credit for finding errors, adding BioNumbers and contributing insights / vignettes.

8 Final project

You'll be presenting a calculation here on the board. You'll have 5 minutes to present. You're also handing in a vignette.

What is a vignette? See the book for examples. Each vignette raises a question (e.g., how fast do cells move?) and aims to provide answers / quantitative context on the question in 2-3 pages.

9 HW 1 & Reading

You will receive **Homework 1** by email. It involves some reading and some calculating about the number of cells in your body. It's due in two weeks. Submission details will be sent to you.

Reading: Introductory sections from the book.

10 Teaser: mRNA vs. Protein

What is more massive: a (spliced) mRNA, or the protein it encodes?

(a) protein
(b) mRNA
(c) Roughly the same (within a factor of 2)
(d) Depends on the gene

11 Handout for class

Darwin on Numeracy

> I have deeply regretted that I did not proceed far enough at least to understand something of the great leading principles of mathematics, for men thus endowed seem to have an extra sense.
>
> Charles Darwin, autobiography

On Being Here Together

> Dear Ron Milo,
> As we near the end of 2016 and look to the New Year, I want to say a heartfelt thank you for being a part of the global edX learner community. Together, we have accomplished so much – 33,601,092 course enrollments, 117,969,427 problem sets attempted, 229,405,155 minutes of video watched in more than 1,300 courses from the world's best universities and institutions.

What do you make of these numbers?

Biological Numeracy Quiz

How large a text could *E. coli*'s replication machinery copy without error?

1. "The Metamorphosis" by F. Kafka ($\approx 10^5$ characters)
2. The Hebrew Bible ($\approx 10^6$ characters)
3. Encyclopedia Britannica ($\approx 10^8$ characters)
4. English Wikipedia ($\approx 3 \times 10^{10}$ characters, see Size of Wikipedia)

Bonus: could you measure this number? Can you do it without sequencing?

Which is more massive – a protein or the mRNA that codes for it?

1. Protein
2. mRNA
3. about the same
4. it depends

Pro sprinters react to a starting gun. What's the fastest reaction time?

1. ≈ 1 ms
2. ≈ 10 ms
3. ≈ 100 ms
4. ≈ 1 s

Vibrio cholera's genome has a 4.03×10^6 base pairs. How many genes?

1. 2200-2800
2. 2800-3600
3. 3600-4400
4. 4400-5000

Bonus: poplar trees genomes have 5×10^8 bp. How many genes do they have?

I engineered *E. coli* to express one copy of p53. What's the p53 concentration?

1. ≈ 1 picomolar (10^{-12} M)
2. ≈ 1 nanomolar (10^{-9} M)
3. ≈ 1 micromolar (10^{-6} M)
4. ≈ 1 millimolar (10^{-3} M)

Bonus: what's the number for a different cell type, e.g., HeLa or B cells.

How much oxygen does an adult human use per minute?

1. a few mL/minute
2. a few tens mL/minute
3. a few hundred mL/minute
4. a few thousand mL/minute

Bonus: How much O_2 does a small houseplant make per minute?

Which is faster, transcription or translation?

1. transcription
2. translation
3. about the same